

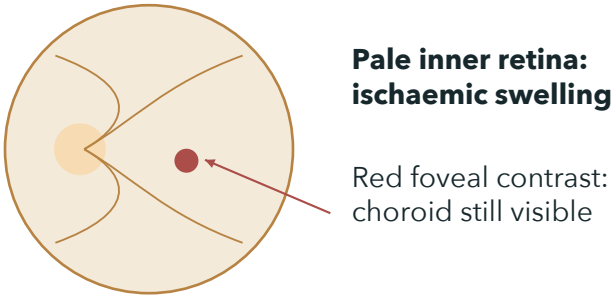
Retinal vascular occlusions

TYPE B | Supply failure versus drainage failure.

Artery: pale ischaemic inner retina

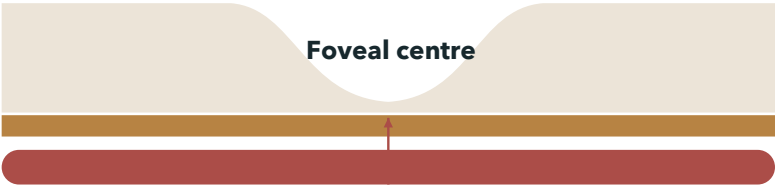
CRAO colour contrast and tissue supply

IDEALISED CRAO FUNDUS



CROSS-SECTION THROUGH FOVEAL CENTRE

Inner retina



Choroid supports outer retina

Thin foveal centre reveals choroidal colour against opaque inner retina. A red centre does not establish preserved central vision.

Read the arterial pattern

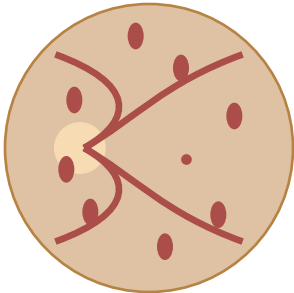
Finding	Mechanistic meaning
Abrupt painless severe loss	Acute retinal afferent injury
Attenuated arteries + pale retina	Arterial supply failure with inner retinal ischaemic swelling
Cherry-red foveal contrast	Thin foveal centre against opaque surrounding inner retina
Sectoral whitening	A branch arterial territory can be involved

Blood supply checkpoint

Central retinal artery: inner retina. Choroid: principally outer retina. Preserved choroidal flow cannot fully rescue an infarcted inner retinal circuit.

Vein: congested and haemorrhagic

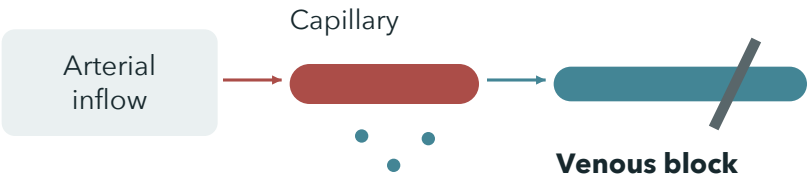
Retinal venous pressure backs up IDEALISED CRVO FUNDUS



Dilated tortuous veins

Haemorrhage in multiple quadrants

Disc swelling; macular fluid may occur



Pressure backs up -> leakage and bleeding

Branch occlusion: one drainage territory

A central venous lesion can affect all four quadrants; a branch lesion follows one drainage territory. These are idealised patterns.

From image to process

Clue	Interpretation
Dilated, tortuous veins	Impaired venous drainage
Flame and dot/blot haemorrhages	Bleeding in superficial and deeper retinal compartments
Macular thickening and fluid	Leakage can reduce central acuity
Extensive non-perfusion	VEGF can drive retinal or anterior neovascularisation

Crossing trap

Chronic hypertensive arteriolar sclerosis can narrow a venule at an arteriovenous crossing: AV nicking. A local crossing change is different from the widespread congestion of CRVO.